



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 • STANDARD 6.GR.4

# Represent Three-Dimensional Figures in Two Dimensions

Lesson 5-6 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can find the surface area of a solid by using its net.

**Language Objective:** I can explain my work using the words surface area, net, face, and two-dimensional.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK – FILL EACH BLANK WITH THE BEST WORD

Surface Area Using Nets

Surface area

Net

Face

Two-dimensional

Edge



Tap any word to see what it means and a picture.

1 The total area of every face, found by adding the faces in a net, is the

2 The total area of all the faces of a solid is its

3 A flat pattern that folds up into a solid is a

4 A flat surface of a solid figure is a

5 A flat shape with length and width but no thickness is

6 A line segment where two faces meet is an

## Write About the Math

**How many sheets of wrapping paper are needed to cover the entire box?**

*¿Cuántas hojas de papel de regalo se necesitan para cubrir toda la caja?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐

**Surface Area Using Nets**

*Área de superficie usando redes*

☐

**Surface area**

*Área de superficie*

☐

**Net**

*Plantilla (desarrollo plano)*

☐

**Face**

*Cara*

☐

**Two-dimensional**

*Bidimensional*

**Model:** *A net lays every face flat so you can see and measure all six faces.* — Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The box's surface area is \_\_\_\_ in<sup>2</sup> because  $SA = 2(\text{____} \times \text{____}) + 2(\text{____} \times \text{____}) + 2(\text{____} \times \text{____})$**   
**= \_\_\_\_ + \_\_\_\_ + \_\_\_\_ = \_\_\_\_.** **You need \_\_\_\_ sheets because \_\_\_\_  $\div$  400  $\approx$  \_\_\_\_.**
- **My answer is \_\_\_\_ because \_\_\_\_.**

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Surface Area Using Nets
2. **Notes 2:** Surface area
3. **Notes 3:** Net
4. **Notes 4:** Face
5. **Notes 5:** Two-dimensional
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is  $9\text{ in} \times 4\text{ in} \times 5\text{ in}$ , a student might add just one face from each pair:  $(9 \times 4) + (9 \times 5) + (4 \times 5) = 36 + 45 + 20 = 101\text{ in}^2$ , but the net actually has two of each face, so the correct total is  $2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202\text{ in}^2$  — exactly double. Before you submit, ask: 'Did I count both matching faces in each pair, or only one from each pair?'*
8. **Try It 1:** A.  $52\text{ in}^2$  — *From the net:  $SA = 2(2 \times 3) + 2(2 \times 4) + 2(3 \times 4) = 12 + 16 + 24 = 52\text{ in}^2$ .*
9. **Try It 2:** — *When a net folds up, each flat shape becomes one face of the solid, so the number and the shapes of the faces identify the solid — six identical squares fold into a cube, while three pairs of matching rectangles fold into a rectangular prism.*